

UTTARADIT, Thailand

WMO: 483510

Lat: 17.6244N

Lon: 100.0969E

Elev: 210

StdP: 14.58

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	60.4	62.9	50.6	55.1	72.6	53.7	61.7	72.8	8.4	81.1	7.2	80.4	0.2	10	0.310

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	20.5	101.6	79.0	99.4	78.8	97.5	78.6	82.0	93.3	81.6	92.5	81.0	91.6	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.5	154.4	86.6	79.0	151.5	86.2	78.4	148.6	85.7	46.1	94.0	45.5	92.8	45.0	91.8	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.1	4.8	DB	56.4	104.3	3.4	2.4	53.9	106.0	51.9	107.4	50.0	108.7	47.6	110.5	
			WB	53.4	84.0	3.5	1.6	50.9	85.2	48.9	86.1	46.9	87.0	44.4	88.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.0	77.2	80.3	84.6	88.2	87.1	85.5	84.2	83.5	83.8	83.3	80.6	77.1
	DBStd	4.57	3.50	3.38	3.35	3.24	3.75	2.53	2.50	2.20	1.94	2.30	3.19	3.99
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	1	0	0	0	0	0	0	0	0	0	0	1
	CDD50	12031	843	849	1073	1147	1151	1066	1061	1039	1015	1031	917	839
	CDD65	6558	379	429	609	697	686	616	596	574	565	566	467	374
	CDH74	74822	3328	4462	7514	9666	8984	7527	6870	6290	6266	6160	4559	3196
(13)	CDH80	35376	1382	2210	4127	5702	4871	3505	2861	2415	2494	2610	1948	1251
(14)	Wind	WSAvg	0.6	0.4	0.6	0.7	0.9	0.7	0.6	0.6	0.4	0.5	0.6	0.6
(15) Precipitation	PrecAvg	58.3	0.4	0.3	1.3	2.6	8.3	8.1	7.4	11.4	11.7	5.5	0.8	0.2
	PrecMax	82.7	4.9	1.5	5.1	7.5	15.7	16.5	13.1	18.9	21.8	9.9	2.8	0.9
	PrecMin	45.0	0.0	0.0	0.0	0.0	2.4	3.0	2.8	4.8	5.6	0.8	0.0	0.0
	PrecStd	9.7	1.2	0.4	1.5	2.4	3.3	4.2	2.5	3.9	4.9	2.7	0.9	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.4	97.3	101.3	104.9	104.8	99.7	97.0	94.9	94.9	95.0	94.8	93.1
		MCWB	73.0	73.8	76.3	79.1	79.5	79.9	80.3	80.9	80.6	79.2	77.5	75.2
	2%	DB	91.4	95.1	98.8	102.2	101.2	96.5	94.5	92.9	93.1	93.5	92.9	90.8
		MCWB	72.8	74.0	76.5	79.2	79.9	80.2	80.2	80.4	80.4	78.9	76.6	74.0
	5%	DB	89.7	93.3	97.0	100.1	98.6	94.3	92.6	91.3	91.7	92.1	91.2	89.1
		MCWB	72.0	74.0	76.5	79.0	80.0	80.2	79.9	80.1	80.2	78.4	75.7	73.1
	10%	DB	87.7	91.4	95.1	97.9	96.0	92.4	90.7	89.6	90.0	90.6	89.4	87.0
		MCWB	71.5	73.6	76.5	78.9	79.8	80.0	79.6	79.8	79.9	78.2	75.2	72.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.2	79.0	80.6	82.6	83.0	82.5	81.9	82.1	82.3	81.8	80.0	78.3
		MCDB	88.1	91.8	95.1	98.4	95.7	93.4	92.3	91.0	91.4	91.3	90.1	88.5
	2%	WB	75.7	77.6	79.5	81.4	81.8	81.6	81.0	81.3	81.5	80.8	79.0	76.5
		MCDB	86.5	90.2	93.9	96.8	94.6	92.1	90.8	90.0	90.2	89.8	88.6	86.2
	5%	WB	74.4	76.4	78.6	80.6	81.1	80.9	80.5	80.7	80.9	80.0	78.0	75.1
		MCDB	85.0	88.9	93.0	95.6	93.5	91.0	89.8	89.2	89.3	88.6	87.0	84.7
	10%	WB	73.1	75.2	77.7	79.9	80.5	80.4	79.9	80.1	80.3	79.3	77.0	73.8
		MCDB	83.4	87.5	91.8	94.5	92.6	90.0	88.6	87.8	88.2	87.4	85.4	83.7
(35) Mean Daily Temperature Range	5% DB	MDBR	22.4	23.9	22.5	20.5	16.7	13.9	12.5	12.3	13.2	15.1	17.9	20.1
		MCDBR	24.6	25.8	24.7	22.9	20.5	16.9	15.7	14.8	15.3	17.0	19.5	21.3
		MCWBR	8.6	9.2	8.2	6.6	5.5	4.8	4.7	5.0	5.1	5.3	6.1	7.1
	5% WB	MCDBR	20.3	22.2	21.9	20.5	17.5	14.9	14.0	13.7	14.2	15.4	17.2	18.7
		MCWBR	7.9	8.9	8.1	6.7	5.7	5.0	4.7	5.1	5.3	5.4	5.8	6.7
(40) Clear-Sky Solar Irradiance	taub		0.468	0.568	0.677	0.638	0.502	0.447	0.437	0.448	0.485	0.539	0.456	0.457
	taud		2.124	1.876	1.656	1.787	2.187	2.368	2.393	2.344	2.225	2.031	2.239	2.179
	Ebn at Noon		257	237	216	226	256	268	271	270	261	241	259	257
	Edh at Noon		47	63	82	72	48	39	38	41	45	54	42	43
(44) All-Sky Solar Radiation	RadAvg		1472	1646	1715	1893	1742	1564	1455	1426	1467	1494	1527	1419
	RadStd		79	69	97	99	145	119	104	84	79	85	77	62

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	+0.52	N/A	N/A	N/A	N/A	+0.22	N/A	N/A	+247	+250

Nomenclature: See separate page